Blind Pedestrian Localization and Reorientation at Urban Crosswalks via UWB device

Rayna Hata[1], Isaac K. Isukapati[2], Zachary B. Rubinstein[2], Stephen F. Smith[2]

***Abstract*— Urban mobility and navigation through signalized intersections pose a challenge for vision impaired pedestrians. Constant vehicle noises, unclear crosswalk guides and cues, and the lack of sufficient time on the signal make crossing these intersections extremely difficult. PedPal, a mobile application, addresses these problems by personalizing the crossing experience for its users through real-time connectivity with the intersection's traffic control system. On one side, connectivity allows the user to communicate crossing intent to the intersection, and on the other PedPal keeps the user informed of the current crossing state. As the user crosses the intersection, PedPal utilizes Bluetooth beacons installed at each corner to track user progress and recognize when the crossing has been completed. Although the localization accuracy of these beacons has proven insufficient for more advanced tracking functions such as monitoring the user's position within the crosswalk, newer Ultra Wide Band (UWB) radio technology offers Centimeter-level accuracy, and provides new opportunities for enhancing the intersection crossing safety features of the PedPal app. In this paper we pursue this possibility. We assume that UWB beacons are installed at each corner of the intersection, and focus on the problem of detecting pedestrian movement outside of the crosswalk. Using distance measurements provided by the UWB beacons over time, we propose a localization and reorienting algorithm to ensure that the pedestrian will get to their destination safe and efficiently. A SUMO simulation model was developed to evaluate both algorithmic components. The localization method was found to provide high accuracy, differing by only 0.003% from the ground truth, and the reorienting algorithm was shown to consistently result in successful crossing within the crosswalk.**

***Index Terms*— Blind-Vision Impaired (BVI), Ultra Wide-Band (UWB),Urban Crosswalks, Blind-Pedestrian Navigation, Path Prediction**

I. Introduction

Walking as a method of transportation is often overlooked as being a challenge, yet for those in the blind/vision-impaired (BVI) community, walking, especially in urban settings, can prove to be a difficult task. A survey sent to 1,123 members of the Association for the Education and Rehabilitation (AER) of Blind's orientation and mobility team found that ninety-eight percent of respondents expressed that difficulty in knowing when to start crossing. Lack of aids such as audio cues or loud background noises interfere with the pedestrian's ability to know when the crossing signal turns green.Furthermore, in the survey, ninety-seven percent of their respondents stated that they had difficulty keeping straight path on a crosswalk. Sixty-six of the respondents of the survey said that they have trouble figuring out where the destination corner is. [1]

Vision-impaired pedestrians employ a range of methods to help cope with the challenges of crossing signalized intersections, including the use of a white cane, a guide dog, and/or cues learned through Orientation and Mobility (O&M) training (e.g., detecting the presence of textured concrete or gauging the sound of oncoming vehicles [2]). However, not everyone has access to a guide dog for reasons such as expenses, allergies, or availability, and the presence of other pedestrians, increased traffic noise, or other unpredictable factors complicate the use of various crossing cues in urban settings.

Additional physical infrastructure is also often added to intersections to provide crossing assistance to vision-impaired pedestrians. Such installments include noise emitting signals that alert the pedestrian when the signal is green and textured curb ramps to direct the pedestrian into the intersection. But, these installments cannot guarantee that the pedestrian will not drift out of the crosswalk as they move across the intersection. Furthermore, both pedestrians and O&M specialists find that looking for the curb ramp can cause more harm than good at times. While looking for the ramp, the pedestrian may get misaligned with the intersection and thus veer off in a different direction when crossing. [3]

Given the limitations of existing methods and infrastructure in supporting pedestrians with disabilities, research has turned attention in recent years to the use of smartphones as an assistive device. Through auditory [4] and haptic [5] feedback, text-to-speech features, and other accessibility features, it is felt that human-computer interactions will allow the vision-impaired community to more effectively and more safely navigate and explore the world. Several recent research efforts have focused specifically on the development of smartphone apps that exploit emerging "connected vehicle" technology to enhance pedestrian safety when crossing intersections. [6], [7], [8]. The *PedPal* safe intersection crossing app [8], [9], for example, has demonstrated that real-time connectivity can be used to personalize the pedestrian's crossing experience, allowing the user to communicate how much time is needed to safely cross to the traffic signal control system, and continuously using information received from the traffic control system to keep the user aware of the current crossing state. It has also demonstrated basic tracking capabilities as the user crosses, which until recently have been limited by localization accuracy issues.

In this paper, we consider the problem of extending

[1] Rayna Hata is with the department of Computer Science, Colby College, Waterville,ME USA rhata23@colby.edu

[2]Isaac Isukapati, Zachary Rubinstein, and Stephen Smith are with the Robotics Institute, Carnegie Mellon University, Pittsburgh, PA, USA (isaack,zbr,sfs)@cs.cmu.edu

PedPal's basic tracking capabilities to enable detection of and reaction to pedestrian movement outside of the crosswalk. Specifically, we propose a method of localizing and tracking pedestrian movement within the crosswalk through the use of ultra wide-band (UWB) radio beacons. UWB signals in outdoor line of sight (LOS) settings have an accuracy up to 2 to 3 cm [10], which offers tremendous sensing advantage over predecessor Bluetooth beacon and smartphone localization technologies, and is sufficient to accurately track a pedestrian. Using multiple UWB beacons positioned at each intersection corner and communicating with the PedPal app, we implement a pedestrian localization algorithm to triangulate the pedestrian's current location as well as determine if they are drifting away from the bounds of the crosswalk. If the algorithm determines that the pedestrian is drifting to the outer bounds, then a reorienting algorithm is applied to guide the pedestrian back to the center of the crosswalk to ensure safe crossing.

II. Background

A. PedPal

As indicated above, the starting point for our research is PedPal [8], [9], a smartphone app designed to promote safe intersection crossing for pedestrians with disabilities. PedPal utilizes emerging "connected vehicle" technology to communicate directly with the traffic control system at the intersection, and to actively influence its behavior to enhance safety. Most basically, PedPal knows the speed at which its user travels (e.g., using a white cane or a walker), and communicates this crossing constraint along with the user's chosen crossing direction to ensure that when the user receives the crossing signal, sufficient time will be allocated for crossing. Reciprocally, the app receives standard Dedicated Short Range Radio (DSRC) MAP and SPaT (Signal Timing and Phase) messages from the intersection to provide crossing options to the user and continual information about the current crossing state to prepare and inform the user about when it is time to cross. PedPal also actively monitors the crossing progress of its user through the intersection, and if it is detected that the user is crossing slower than expected (e.g., stumbles in a pothole and is slow getting up) then PedPal will request that the traffic signal system dynamically extend the green time to compensate. Pedpal also broadcasts its user's presence in the crosswalk to any approaching connected vehicles. Finally, for certain disabled pedestrians (such as motorized wheelchair users) PedPal can communicate expected arrival time information to the next intersection for purposes of streamlining intersection crossing time.

PedPal's functional capabilities are achieved by integrating the app with two distinct infrastructure technologies. First, PedPal is designed to inter-operate with Surtrac, a decentralized, real-time adaptive traffic signal control system [11], [12]. Within a Surtrac deployment, a processor is placed at each signalized intersection. At each intersection, a new traffic signal timing plan that minimizes the cumulative delay of all currently sensed traffic approaching the intersection is computed in real-time. Whenever a new timing plan is generated, the intersection then communicates to its downstream neighbors what traffic it expects to be sending their way (according to the plan), providing a basis for network-level coordination. Surtrac has been shown to lower average vehicle travel times by 25%, average number of stops by 30% and average amount of idle time by 40%. [11]. Surtrac is designed to incorporate multi-modal traffic flows, and as such, it is straightforward to factor pedestrian crossing demands into the signal timing plans that are generated.

The above mentioned capabilities provided by PedPal rely rather heavily on an ability to accurately determine when user's have arrived at a given intersection corner. In this regard, the native localization capability of the smartphone (in this case an Apple iPhone) has proved to be less than adequate. This has led to introduction of a second infrastructure technology, a set of fixed-position Bluetooth beacons located at each corner of the intersection. With this addition, it has been possible to use geo-fencing to provide the basic ability to recognize corners, and, for short crossing segments, this translates into some ability to track user crossing progress.

B. Movement to UWB beacons

The advent of UWB technology, however, has fundamentally changed the localization landscape. Some early experimentation with Apple AirTag technology indicated the potential for order of magnitude improvement in the ability to accurately determine object distances from the phone app, and more recent work with PedPal has utilized UWB technology to dramatically improve detection of arrival intersection corners and estimation of progress made toward crossing a particular street segment. For our initial experimentation in the field, a set of 4 Apple iPhones (Model 11 and above) have been utilized as initial UAB corner"beacons" at the intersection of Highland Avenue and Centre Avenue in the east end of Pittsburgh PA (see Figure 2). The ability to continuously exchange distance information between these UWB corner beacons and the PedPal app was established using Apple's Nearby Interaction framework[1], and initial experimentation has shown the ability to detect both arrival at intersection corners and the completion of crossing trips with a few centimeter accuracy up to a distance of 60 feet (the longest leg of the test intersections. Next steps will be to explore incorporation of 3rd party UAB beacons that are emerging on the market and support Nearby Interaction.

III. Detecting and Reacting to Movement Outside of the Crosswalk

Given the observed localization accuracy that is achievable with the use of UWB corner beacons, we consider their use in providing a more advanced tracking capability that relates directly to pedestrian safety, that of detecting and responding to pedestrian movement outside of the crosswalk. We can expect the MAP message received by the PedPal app from the intersection to provide the precise coordinates

[1] https://developer.apple.com/documentation/nearbyinteraction

of the crosswalk to be traversed. Thus, the challenge is one of situating the pedestrian within this physical space and prediction of the pedestrian's current trajectory. Accordingly, our solution is broken into two algorithmic components. Algorithm 1 involves localization of the pedestrian within the crosswalk. Algorithm 2 then predicts the pedestrian's future path, and, if it is determined to be drifting out of the crosswalk, the pedestrian is alerted to take corrective action.

Algorithm 1: Pedestrian Localization Process

$\vec{A} \leftarrow$ vector representing Geo locations of anchors
$d'_{i,\rho} \leftarrow$ distance between pedestrian and anchor i
$P'_{i,\rho} \leftarrow$ predicted Geo location of pedestrian w.r.t anchor i
$\vec{P'} \leftarrow$ vector representing predicted geo-locations of pedestrian from each anchor
while *The Crossing is incomplete* **do**
  Compute $\angle A$, $\angle B$ using $d, d_{1,\rho}, d_{2,\rho}$
  Estimate $P'_{i,\rho}$ from each anchor
  Set $\hat{P}$ as average of $P'_{1,\rho}$ & $P'_{2,\rho}$
  Call Path Prediction and Reorientation
end

Given a crossing episode between corner 1 and corner 2, Algorithm 1 periodically receives $d'_{1,\rho}$ and $d'_{2,\rho}$, the current distance readings between the pedestrian and each anchor respectively, throughout the pedestrian crossing. These distances along with the distance between the beacons (d) are used to geo-locate pedestrian ($P'_{i,\rho}$) with respect to each anchor. These calculations can be simply carried out using cosine rule as shown in Figure 1:

$$\cos(A) = (d^2 + {d'_{1,\rho}}^2 - {d'_{2,\rho}}^2)/(2 * d'_{1,\rho} * d'_{2,\rho}) \quad (1)$$

$$\cos(B) = (d^2 + {d'_{2,\rho}}^2 - {d'_{1,\rho}}^2)/(2 * d'_{1,\rho} * d'_{2,\rho}) \quad (2)$$

The final geo-location of the pedestrian is computed as the average of ($P'_{1,\rho}$, $P'_{2,\rho}$).

Next, we forecast pedestrians future path in order to determine if they are drifting out of the crosswalk. The heart of this algorithm is to determine and react quickly to the possibility of the pedestrian drifting out into the intersection. In order to do so, we must estimate and predict the next movements with reference to their initial pose estimate.

The pedestrian's initial pose estimation $\vec{O}^0_\rho$ is set as the reference coordinate against the succeeding pose estimations $\vec{O}^k_\rho$ to obtain the slope $m^j_{\rho,k}$ at step k. Step k indicates the number of readings taken in since the pedestrian started the crossing. The slope indicates the orientation that the pedestrian walks in relative to their original starting point at the beginning of the crossing.

If there is a change in the slope, it can be assumed that there is a change in the direction at which the person is walking in. However, it is necessary to compile multiple slope calculations to accurately predict if the pedestrian is drifting towards the outer bounds of the crosswalk. Hence we use a heap queue Q to store j consecutive slope calculations. By taking the median value of Q,

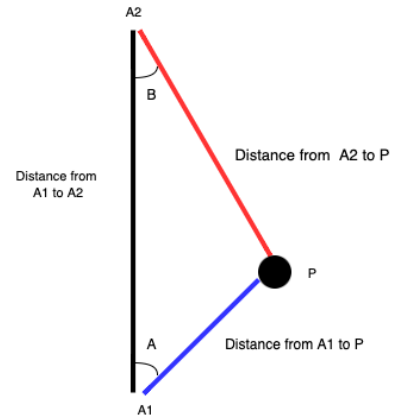

Fig. 1. : This is a geometrical representation of the anchor and the pedestrian's location. The black line is the distance from anchor 1 to anchor 2. The blue line shows the distance between anchor 1 and the pedestrian, and the blue line shows the distance between anchor 2 and the pedestrian. In order to predict the pedestrian's location, we use angle 1 and angle 2 in equation 1 and 2.

The median taken after collecting j slope calculations is subsequently used as a reference slope for determining if the pedestrian is straying. Thus, at each step k after the first j points, the oldest median is popped out of the queue and the new slope between the pedestrian's initial pose estimation and the current pose estimation is added to the heap queue. After the slope is added, we compute the new median of the queue and compare that with the original median. It is important to use the reference slope as using any other slope will determine the pedestrian's direction from that slope.

If the new median is outside the threshold bounds, then there is indication that the pedestrian is not walking in a straight line towards the end of the crosswalk, and can no longer safely complete the crossing. If this movement continues, there is a high probability that pedestrian will leave the bounds of the crosswalk.

However, it is critical not to give instructions to the pedestrian until we can be sure that an outlier did not cause the reading. Therefore, we will continue to observe and not interfere until the pedestrian's slope is outside of the threshold for more than τ time steps. If that is true, then we start instructing the pedestrian to turn in the opposite direction of the slope. If the current median is larger than the threshold, then we will instruct the pedestrian to turn right. If the median is less than the threshold, then we will instruct the pedestrian to turn left.

The algorithm continues to localize and monitor the orientation of the pedestrian until the crossing has been completed. However, once the slope is within the threshold again, the algorithm determines that the pedestrian is back in the bounds of the crosswalk, and thus will stop giving instructions.

Algorithm 2: Path Prediction and Reorientation Process

$\vec{O}_\rho^0 \leftarrow$ pedestrian's initial pose estimation
$\vec{O}_\rho^k \leftarrow$ pedestrian's pose estimation at step k
$m_k \leftarrow$ slope of $(\vec{O}_\rho^k, \vec{O}_\rho^0)$
$Q \leftarrow$ heap queue containing slopes of last j steps
$m_\rho^0 \leftarrow$ initial pedestrian orientation (median of Q after first j steps)
$m_{\rho,k}^j \leftarrow$ pedestrian orientation at step k based on last j steps
$\vec{M}_t \leftarrow$ threshold bounds of median
$\tau \leftarrow$ consecutive number of steps $m_{\rho,k}^j$ is outside threshold bounds
$\tau_{max} \leftarrow$ max allowed number of steps for $m_{\rho,k}^j$ to be outside threshold bounds
At each step k do:
compute m_k & update Q
set median of Q as $m_{\rho,k}^j$
if $m_{\rho,k}^j \notin \vec{M}_t$ **then**
  τ++
end
if $\tau == \tau_{max}$ **then**
  start correction
end

IV. Evaluation

A. SUMO Simulation Framework

In order to visualize the pedestrian's movements and accurately reflect traffic conditions, the localization and path-planning algorithm is implemented in the Simulation of Urban Mobility (SUMO) software. SUMO is an open source continuous traffic simulation package that allows for the inter-modal simulation of urban traffic that displays the movements of vehicles and pedestrians on road structures imported via open-street maps. To simulate an intersection where the SURTRAC system is already implemented, we imported the intersection of South Highland Ave. and Centre Ave in Pittsburgh, PA (see Figure 2).

Fig. 2. : A google earth image of Centre Ave. and Highland Ave., Pittsburgh,PA. In this image, Centre Ave is the horizontal line and Highland Ave. is the vertical line.

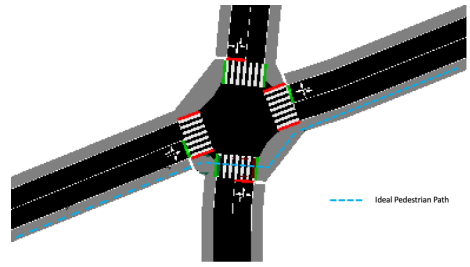

Fig. 3. : SUMO Visual of Pedestrian Path. The dotted line indicates the ideal pedestrian path.

Traffic Control Interface (TracI) allows for manipulating such pedestrians and vehicles as the simulation progresses. Using TracI, scenarios of the pedestrian drifting to the edge of the crosswalk are tested to mimic the possible movements of a blind pedestrian. As mentioned in the introduction, multiple factors, such as a reaction to traffic noise or an incorrect setup at the start of the intersection, can cause those in the BVI community to drift towards the edge of the crosswalk.

To manipulate and estimate the pedestrian's coordinates, we take in and control several parameters outputted by SUMO. Such parameters are pedestrian coordinates in (x,y) format, angle that the pedestrian is facing, speed, and the time step of the simulation advancing in 100-millisecond intervals. Each of these parameters reflect data that would taken in by PedPal.

Another data point that PedPal would be receiving from the UWB devices is the distances from each anchor. As UWB device functions do not exist in SUMO, we must manually compute what the smartphone would be seeing as distances from the UWB anchors. This is done so by finding three euclidean distances:

- $d_{1,\rho}$= Euclidean distance between anchor 1 and pedestrian
- $d_{2,\rho}$ = Euclidean distance between anchor 2 and pedestrian
- d = Euclidean distance between the UWB anchors

However, these distances have no signal noise and are thus not a realistic representative of readings given in outdoor settings. Therefore, each distance between the anchors and the pedestrian is put through a noise model to emulate a real-world scenario.

The noise level added to the actual distance will increase or decrease as the pedestrian progresses along the intersection. This is reflective of the increase and decrease in noise as the smartphone is further away or closer to the UWB device. If the actual distance is more than half the length of the Euclidean distance between the two anchors, then a random δ_s of a range between 0.05-0.1m is added. If the distance is less, then a random δ_s of a range between 0.0-0.05m is added. The algorithm will take in the following new distances as the actual measurement between the pedestrian and the anchors, which is the equivalent of what the smartphone would be receiving as the distances between the pedestrian

and the anchor in :

- $d'_{1,\rho}$ = Euclidean distance between anchor 1 and pedestrian
- $d'_{2,\rho}$ = Euclidean distance between anchor 2 and pedestrian

In the scenario used in this simulation to evaluate the algorithm, the pedestrian starts the crossing in the correct direction but drifts towards the upper bounds of the crosswalk. This is used to emulate a scenario in which a factor such as noise or other pedestrians causes the person to start turning in the wrong direction. We determined the algorithm a success if it was able to do the following:

- Accurately predict the pedestrian's location with respect to the ground truth
- Detect when the pedestrian started to drift
- Reorient the pedestrian in the correct direction

B. Results

Using the localization method, it can be seen that the localization estimation of the pedestrian's location is on target with the ground truth. Figure 4 shows the plot of each of the pedestrian location estimations. The top-left has the pedestrian's ground truth locations plotted in blue. The bottom-left is the plot of the average of estimated X, Y coordinates from A_1 and A_2 plotted in yellow, and the ground truth coordinates are plotted in blue. The top-right shows the estimated X and Y coordinates with respect to the A_2 anchor in purple with the ground truth in blue. Finally, the bottom-right shows the estimated X and Y coordinates with respect to A_1 in orange and the ground truth in blue. The estimated X coordinate showed an average of 0.018% error from the ground truth, and the estimated Y coordinate showed an average of 0.027% error from the ground truth.

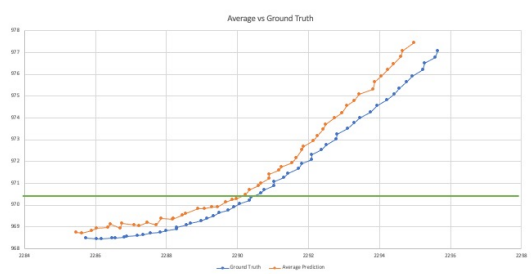

Fig. 4. : Plotting of the Coordinates. In this figure, the blue line shows the plot of the X,Y coordinates for the ground truth and the orange line shows the predicted X,Y coordinate of the pedestrian using the average of the two coordinate points from anchor 1 and anchor 2. This is plot shows the pedestrian's trip before the reorienting algorithm is applied. The green line indicates the upper bound of the crosswalk.

Applying the path predictor algorithm to the pedestrian's movement, we successfully predicted when the pedestrian was drifting from the bounds of the crosswalk and reoriented them to go back to the middle. When the pedestrian is drifting towards the upper bounds of the crosswalk, the algorithm will reorient the pedestrian by telling them to turn right and keep orienting them until the pedestrian reaches an acceptable movement slope relative to the original slope. When the pedestrian is below the acceptable slope threshold, the algorithm urges them to reorient themselves by turning to the left. This test was done on the pedestrian drifting to the lower bounds of the crosswalk as well.

Figures five and six represent two tests we ran on the algorithm. In figure five, the blue path represents the complete trip of the pedestrian before the correction algorithm is run. In this scenario, the pedestrian is drifting towards the lower bound, and without correction, they will leave the crosswalk. The yellow line shows the pedestrian's path when the path prediction and reorientation are applied to their walk. Each point on the line indicates a step that the pedestrian takes. The yellow and blue lines follow the same path until the algorithm shows that the pedestrian is drifting to the lower bounds of the crosswalk. Then, it reorients the pedestrian and allows for a successful crossing.

Figure two represents a scenario where the pedestrian drifts towards the crosswalk's upper bound. Without the course correction and prediction, the pedestrian leaves the bounds of the intersection and has the potential to walk into a vehicle's way. The path prediction algorithm can accurately predict that the pedestrian is slowly straying off the crosswalk path and can stop that promptly. The figure shows that the yellow and blue points are the same until the algorithm determines that the pedestrian is drifting too far. Upon recognition, the algorithm successfully reorients the pedestrian back to the inner bounds of the crosswalk.

Fig. 5. :Recognition and Correction of pedestrian drifting down.

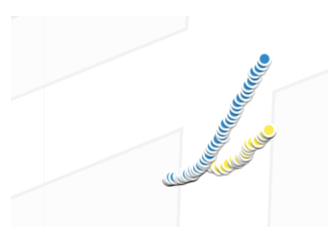

Fig. 6. : Recognition and Correction of pedestrian drifting up.

V. Related Works

A. GPS tracking

GPS tracking is a common alternative to Bluetooth and UWB tracking. GPS tracking can potentially address the

issue of determining if the pedestrian is out of the crosswalk's bounds. In one study, the GPS is used to access the pedestrian's coordinates, and a satellite image of the location is pulled into a crosswalk detection framework to guide the user toward the intersection. While this method is helpful for real-time positioning of the pedestrian's location, the paper mentions that due to factors such as image acquisition problems or GPS accuracy, the localization accuracy cannot be narrowed down to cm. [13] Furthermore, in this method, there is no communication between the mobile device and the signal. Therefore, while the satellite can guide the pedestrian to the desired crosswalk, it cannot notify the pedestrian of the state of the crosswalk, nor can it guide them during the crossing.

B. Indoor UWB Localization

UWB localization can be used in various settings. For example, UWB is used to track unmanned aerial vehicles [14], objects in a factory environment to increase automation and efficiency [15], or in indoor pedestrian tracking systems. One proposed method tracked a pedestrian in an indoor setting by mounting a fusion of UWB sensors and inertial measurement units (IMU) sensors on the foot to track the pedestrian [16]. This method allows for error correction of the tracking measurements caused by non-line of sight instances. While this method effectively tracks the location of the pedestrian, this has not been tested in outdoor environments. Furthermore, the number of sensors and the need to be placed on the foot do not allow for seamless integration in the pedestrian's life.

C. Smartphone Based Tracking

Smartphones as assistive devices have increased over the years as smartphone capabilities advanced. Applications such as BlindHelper [17] utilize smartphones to help blind pedestrians navigate their surroundings. BlindHelper is a smartphone-based application that takes location information, and traffic light status from the phone using GPS signals and outputs commands for the pedestrians to follow. The phone is connected via Bluetooth to a keypad that allows pedestrians to easily select their routes and a sonar distance meter to detect obstacles. This device allows for accurate tracking with an accuracy of up to 0.11m. However, the BlindHelper requires the pedestrian to hold several devices in their hand at once. This can often be inconvenient as their hands will not be free to carry their belongings or hold anything.

D. Pedestrian to Infrastructure Communication

Pedestrian-to-infrastructure (P2I) is a crucial way to allow for communication between the smartphone and the signal. Smart Walk Assistant (SWA) allows for WiFi communication between a smartphone and a roadside unit. The P2I connection implemented in SWA enables users to send a pedestrian signal request to the traffic signal controller and receive the traffic signal status. SWA also has a pedestrian-to-vehicle (P2V) communication that sends the location, speed, and heading of a vehicle to predict any conflicts between the pedestrian and the vehicle. [7]

VI. Conclusion and Next Steps

Precise localization is crucial when giving direction to those with visual impairments to obtain their trust in the assistive device. PedPal's goals are to localize the pedestrian, predict the path, and quickly respond to the pedestrian's drift with the accuracy needed in such devices.

The field tests done on the UWB devices in outdoor settings show that UWB devices provide a reliable and accurate connection between the signal and the smartphone. Evaluating the initial implementation of UWB devices with the SUMO simulation environment, results show that using the distances from the UWB sensors to the pedestrian can accurately localize the pedestrian and follow their path. When compared to the ground truth coordinates measured in the simulation, the localization coordinate predictions show a minimal error. Thus we can conclude that using UWB devices fits the needs of accuracy in PedPal when localizing the pedestrian.

Testing the path prediction and reorientation algorithm in the SUMO simulation environment, we found that the algorithm can successfully predict when the pedestrian is leaving the bounds of the crosswalk. Then the algorithm can respond promptly to reorient the pedestrian without overwhelming the pedestrian with directions.

The next steps include phasing out the simulation setting and testing at real-world intersections. Furthermore, the current simulation moves the pedestrian, assuming they will start to drift out of the crosswalk in the middle of their trip. That is not the only way that can cause a pedestrian to leave the bounds of the crosswalk. We want to implement scenarios where pedestrians may start crossing the intersection already facing the wrong direction. In order to do so, we will need to incorporate the inertial measurement unit sensor's data into the pedestrian path prediction to understand their original orientation.

VII. Acknowledgements

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